**25900902** **Reg.No : .....**

**Name : .....**

**MAHATMA GANDHI UNIVERSITY, KOTTAYAM**

**MGU-UGP (HONOURS) REGULAR /IMPROVEMENT/ REAPPEARANCE**

**EXAMINATION NOVEMBER 2025**

**FIRST SEMESTER**

**Discipline Specific Core Course (DSC) - MG1DSCMOS100 - PC HARDWARE AND SMARTPHONE TROUBLESHOOTING**

(2024 ADMISION ONWARDS)

**Duration: 1.5 Hours** **Maximum Marks: 50**

***Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)***

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

**Part A**

Multiple Choice Questions

Answer **All** Questions

Each Question carries **1** mark

1. Expansion slots are used for which purpose? [U] / [CO1]

   a). Cooling system b). Connect power cables

   c). Install expansion cards d). Store data

2. Which problem in SMPS can cause frequent restarting of a computer [U] / [CO2]

   a). Unstable power supply or faulty capacitors b). Malware infection

   c). Incorrect BIOS configuration d). Broken Ethernet cable

3. Why are GPS antennas designed with high sensitivity in smartphones? [U] / [CO3]

   a). To avoid interference with other sensors b). To improve the speed of internet browsing

   c). To enable faster charging of the device d). To accurately detect weak satellite signals for precise location services

4. What is the primary function of a UPS? [U] / [CO1]

a). Convert AC to DC power

b). Provide continuous power during outages

c). Store energy for future use

d). Improve the efficiency of electrical devices

5. Why is it important to follow a logical flowchart in troubleshooting rather than jumping straight to a solution? [U] / [CO2]

a). To ensure all team members get involved

b). To minimize the risk of solving the wrong problem

c). To create an opportunity to gather more data

d). To avoid creating new problems

6. What is a disadvantage of an offline UPS compared to an online UPS? [U] / [CO1]

a). Higher efficiency

b). Longer transfer time

c). Lower cost

d). Simplicity

7. What is the purpose of the flash memory chip on a smartphone motherboard? [A] / [CO3]

a). To provide temporary storage for the CPU

b). To permanently store the operating system and user data

c). To control the smartphone's wireless communications

d). To regulate the battery charging process

8. Which type of problem might arise from a poorly seated memory module in its slot? [U] / [CO2]

a). High temperature

b). Failure to boot or system crashes

c). Increased power consumption

d). Reduced graphics quality

9. What is the primary function of the BIOS? [U] / [CO2]

a). To load the operating system

b). To manage system memory

c). To provide an interface for application software

d). To perform hardware initialization and provide runtime services

10. When a smartphone cannot connect to a Wi-Fi network, which part should be investigated first? [A] / [CO3]

a). Battery

b). RF Section

c). CPU

d). GPU

**[10x1 = 10]**

## Part B

Short answer Type Questions

Answer any **4** Questions

Each Question carries **5** marks

| | | |
|---|---|---|
| 11. | Explain the role of the Audio IC in a smartphone and how it ensures high-quality sound. | [U] / [CO3] |
| 12. | What type of connector is typically used to attach audio jacks to the motherboard? | [U] / [CO1] |
| 13. | Explain the Power-On Self-Test (POST) process and its significance in the boot sequence of a computer. | [U] / [CO2] |
| 14. | What is the primary function of storage in a smartphone, and what types of memory are commonly used? | [K] / [CO3] |
| 15. | Discuss the different types of RAM and their significance in a computer system. | [U] / [CO1] |
| 16. | What are the safety precautions to be followed while testing SMPS voltage using a multimeter? | [U] / [CO2] |

**[4x5=20]**

## Part C

Essay Type Questions

Answer any **2** Questions

Each Question carries **10** marks

| | | |
|---|---|---|
| 17. | Explain how Recovery Mode can assist in resolving software issues on a smartphone. Provide examples of tasks that can be performed in this mode. | [U] / [CO3] |
| 18. | Explain the function of an SMPS in a computer system. Discuss the significance of the different wire colors in the ATX power supply connector, and how they correspond to specific voltage outputs and connections. | [U] / [CO1] |
| 19. | What is personal computer? What are the different generations of computer? Explain briefly. | [K] / [CO1] |
| 20. | Examine the challenges that users face when interpreting POST indications and the implications for hardware troubleshooting. | [QNSMARTI:T:B20] / [CO2] |

**[2x10 =20]**